Rear Output

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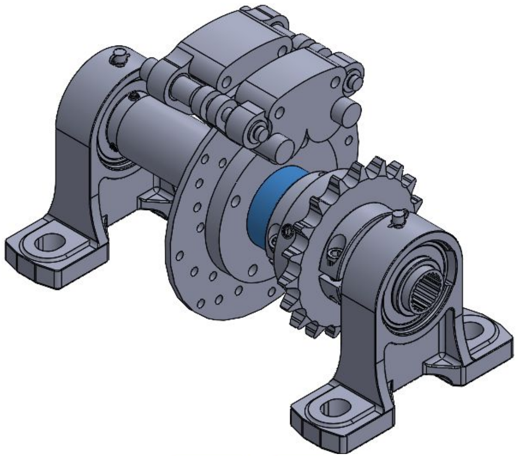
Intro to Rear Output

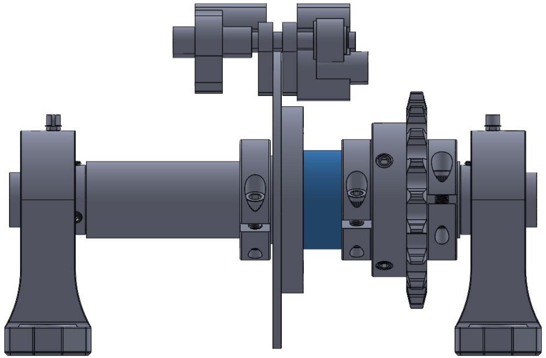
The Goal of Rear Output:

- Transfer **power** from the engine through a gearbox to the back wheels and to the front spool

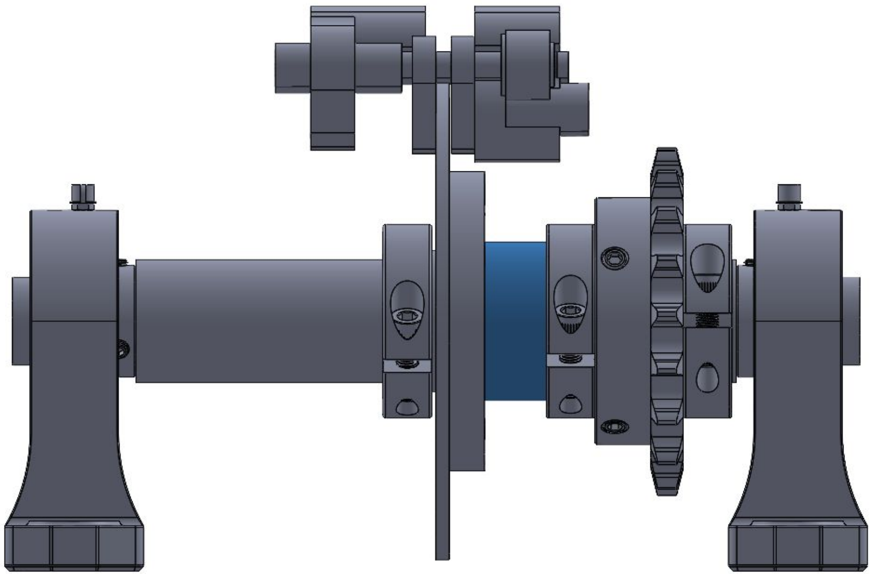


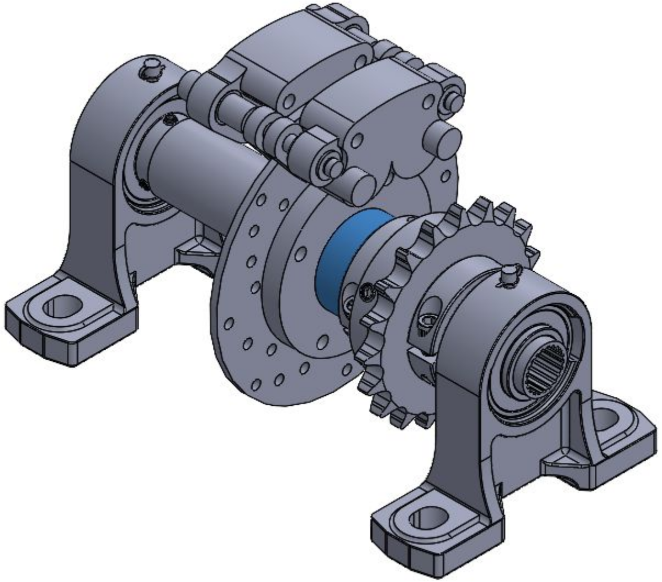
Important components of the Rear Output:

- Rear Spool
- Splines
- Sprockets
- Bearing and Bearing mount
- Brake Rotor

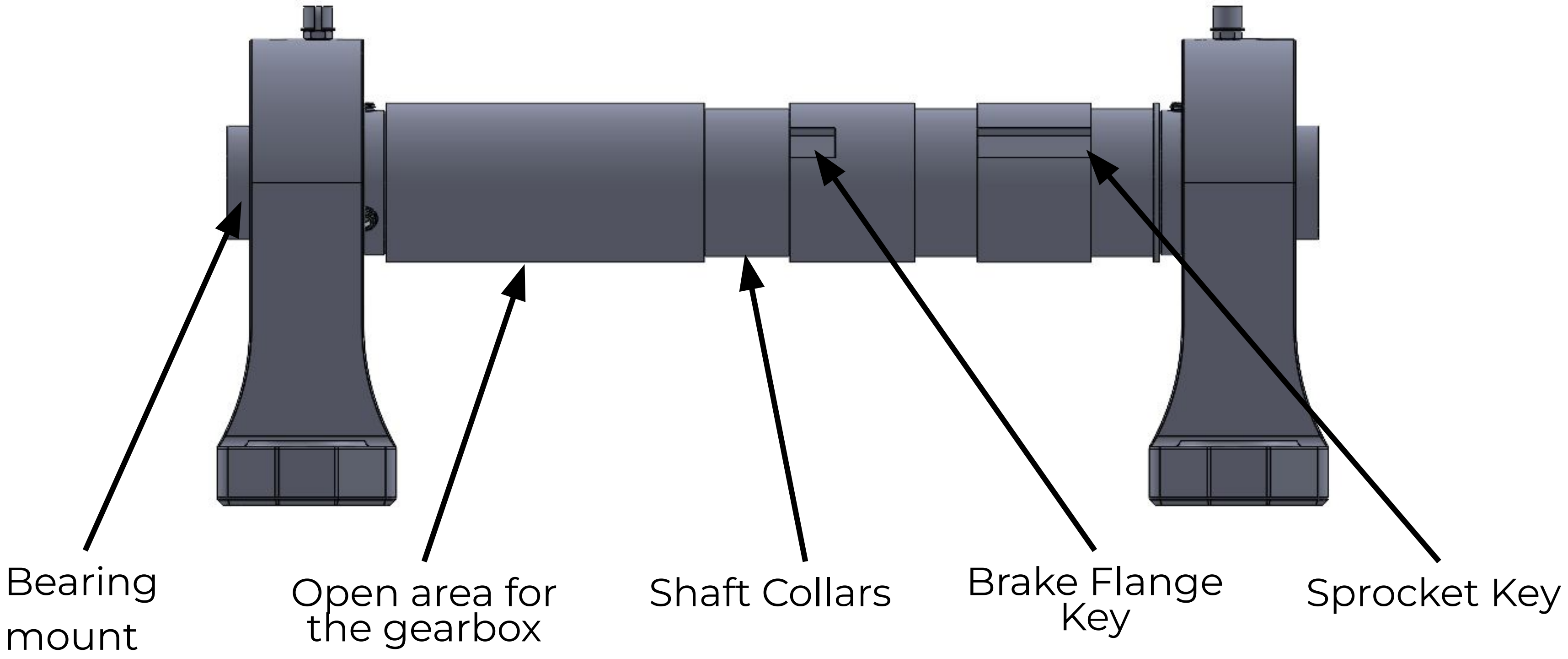


CAD of the Rear Output

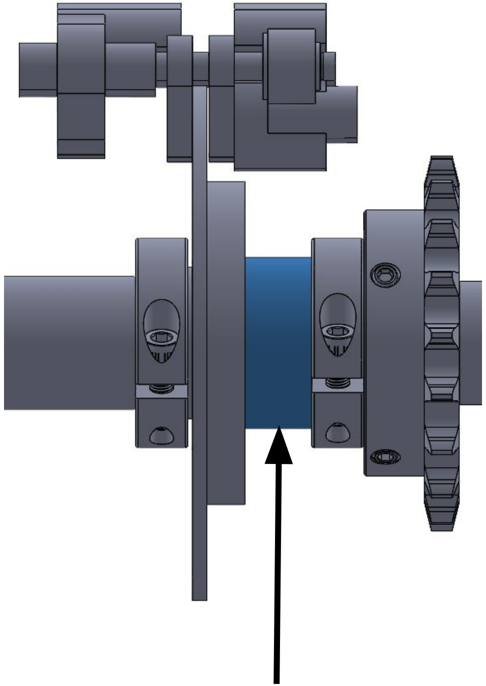




Rear Spool

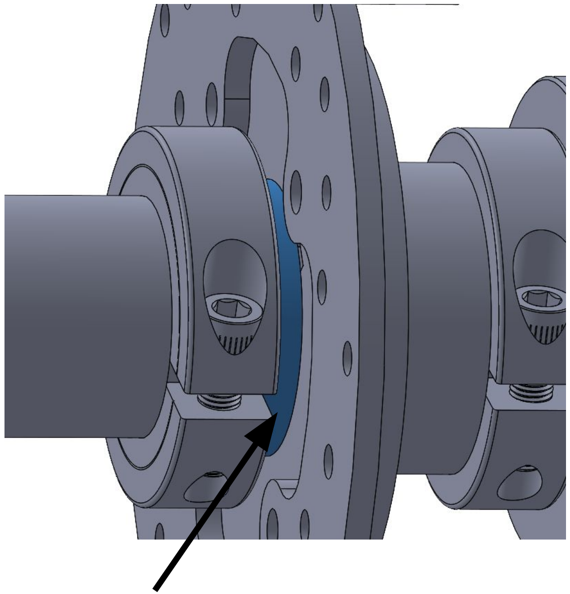

Bearing mount
Open area for the gearbox
Shaft Collars
Brake Flange Key
Sprocket Key

Rear Spool- spacers



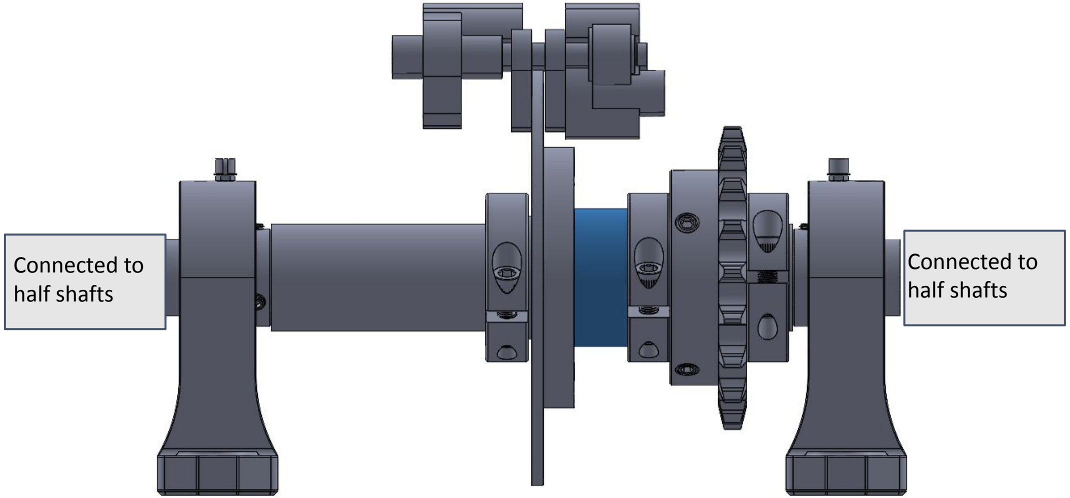
spacer

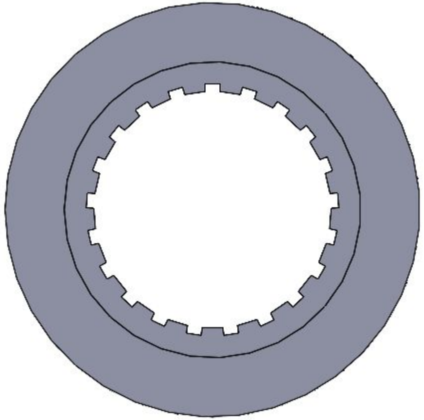
We decided to cad spacers in order to first clear the brake calipers and also to make sure we can access the shaft collars.



spacer

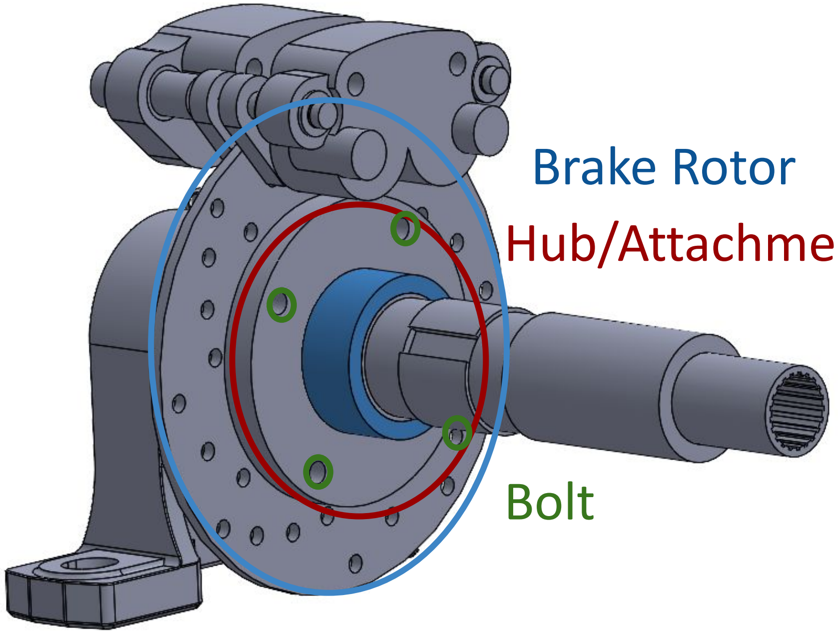
Splines

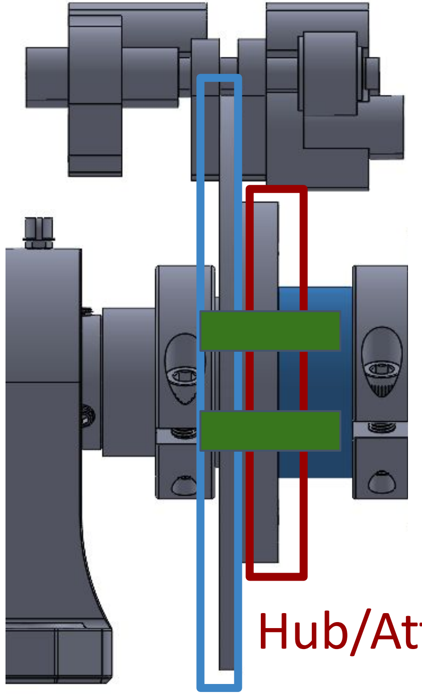




Broaching the spool in order to connect to the half shaft

Parts of Brake





Purpose of Code- Brake

- **Main Purpose: Check to see if the bolts will shear when the rotor stops due to brakes**

```
clear

% Amonton's Law or Coulomb's Law of Friction
% Friction Force (Newtons) = Coefficient of friction * Normal Force (Newtons)

friction_coefficient = 0.42; % Friction Coefficient (hard steel on hard steel),
% https://engineeringlibrary.org/reference/coefficient-of-friction
surface_area = 0.067; % Surface Area (in^2) - area of contact,
% https://www.engineersedge.com/fastener_thread_stress_area.htm
clamp_force = 7000; % Clamping force (lbs) -
% https://sabreindustrial.com/content/Tech%20PDFS/003%20Bolt%20Clamp%20Load%20Chart.pdf
torque = 1200; % Break torque (Nm) for the system -
% https://www.ijert.org/research/braking-calculation-for-a-baja-atv-IJERTV10IS040079.pdf
radius = 0.045212; % Radius (meters) - distance from the center to the bolt where friction acts

% Convert clamping force to Newtons (from pounds-force)
normal_force = clamp_force * 4.4482216153; % Pounds-force to Newtons
% Convert area from square inches to square meters
surface_area = surface_area * 0.00064516; % Conversion from in^2 to m^2
% Calculate the friction force for one bolt
friction_force = friction_coefficient * normal_force;
% Multiply by 4 for the total friction force from all bolts
total_friction_force = friction_force * 4;
% Calculate the frictional torque: Friction force * radius
friction_torque = total_friction_force * radius;

factor_of_safety = friction_torque/torque


disp(['Applied torque: ', num2str(torque), ' Nm']);
disp(['Friction torque: ', num2str(friction_torque), ' Nm']);
% Comparison: If the frictional torque is greater than the applied torque
if friction_torque > torque
    disp('The friction force is GREATER than the torque. The bolts can hold.');
elseif friction_torque < torque
    disp('The friction force is LESS than the torque. The bolts may shear.');
else
    disp('The friction force is EQUAL to the torque.');
end
```

How Code Works- Brakes

Amonton’s Law/Coulomb’s Law of Friction=

Friction Force= *Coefficient of Friction * Normal Force*

Friction Force Total = *Friction Force* * 4 bolts

Frictional Torque = *Friction Force Total* * Radius (1.78 in or 0.045212 m)

We want the *Frictional Torque* > Applied Torque so that it doesn't break

Coefficient of Friction-Brakes

We're using 4130 steel on 4130 steel

We want the highest friction value to optimize

[Coefficient of Friction | Engineering Library]

Table IV: Coefficients of Static and Sliding Friction

Materials	Static		Sliding	
	Dry	Greasy	Dry	Greasy
Hard steel on hard steel	0.78	0.11 (a)	0.42	0.029 (k)
	---	0.23 (b)	---	0.081 (e)
	---	0.15 (c)	---	0.080 (i)
	---	0.11 (d)	---	0.058 (j)
	---	0.0075 (p)	---	0.084 (d)
	---	0.0052 (h)	---	0.105 (k)
	---	---	---	0.096 (l)
	---	---	---	0.108 (m)
	---	---	---	0.12 (a)
Mild steel on mild steel	0.74	---	0.57	0.09 (a)

Unknowns & Assumptions- Brakes

Applied Force = 1200 N

Thus this force required on each wheel is calculated using the "Second Law of Motion". The braking force at each front wheel ("F_{bf}") can be calculated as,

$$F_{bf} = (DF_{f\,mass} * a_{rtd})/2 \quad (9)$$
$$= (167.22 * 13.7146)/2$$
$$F_{bf} = 1146.677 \text{ N}$$

[Braking Calculation for a BAJA ATV]

Clamp Force = Worst case SAE grade 8 = 7000 lb

Suggested Tightening Torque1 Values To Produce Corresponding Bolt Clamping Loads

Size	Bolt Dia. D (in.)	Tensile Stress Area A (sq. in.)	SAE Grade 2 Bolts: Tensile Strength (min psi)	SAE Grade 2 Bolts: Proof Load (psi)	SAE Grade 2 Bolts: Clamp[2] Load P (lb.)	SAE Grade 2 Bolts: Tightening Torque Dry K=0.20	SAE Grade 2 Bolts: Tightening Torque Lub. K=0.15	SAE Grade 5 Bolts: Tensile Strength (min psi)	SAE Grade 5 Bolts: Proof Load (psi)	SAE Grade 5 Bolts: Clamp[2] Load P (lb.)	SAE Grade 5 Bolts: Tightening Torque Dry K=0.20	SAE Grade 5 Bolts: Tightening Torque Lub. K=0.15	SAE Grade 7[3]: Clamp[2] Load P (lb.)	SAE Grade 7[3]: Tightening Torque Dry K=0.20	SAE Grade 7[3]: Tightening Torque Lub. K=0.15	SAE Grade 8[4]: Clamp[2] Load P (lb.)	SAE Grade 8[4]: Tightening Torque Dry K=0.20	SAE Grade 8[4]: Tightening Torque Lub. K=0.15
						lb. in.	lb. in.				lb. in.	lb. in.		lb. in.	lb. in.		lb. in.	lb. in.
4-40	0.1120	0.00604	74,000	55,000	240	5	4	120,000	85,000	380	8	6	480	11	8	540	12	9
4-48	0.1120	0.00661			280	6	5			420	9	7	520	12	9	600	13	10
6-32	0.1380	0.00909			380	10	8			580	16	12	720	20	15	820	23	17
6-40	0.1380	0.01015			420	12	9			640	18	13	800	22	17	920	25	19
8-32	0.1640	0.01400			580	19	14			900	30	22	1100	36	27	1260	41	31
8-36	0.1640	0.01474			600	20	15			940	31	23	1160	38	29	1320	43	32
10-24	0.1900	0.01750			720	27	21			1120	43	32	1380	52	39	1580	60	45
10-32	0.1900	0.02000			820	31	23			1285	49	36	1580	60	45	1800	68	51
1/4-20	0.2500	0.0318			1320	66	49			2020	96	75	2500	120	96	2860	144	108
1/4-28	0.2500	0.0364			1500	76	56			2320	120	86	2860	144	108	3280	168	120
						lb. ft.	lb. ft.				lb. ft.	lb. ft.		lb. ft.	lb. ft.		lb. ft.	lb. ft.
5/16-18	0.3125	0.0524			2160	11	8			3340	17	13	4120	21	16	4720	25	18
5/16-24	0.3125	0.0580			2400	12	9			3700	19	14	4560	24	18	5220	25	20
3/8-16	0.3750	0.0775			3200	20	15			4940	30	23	6100	40	30	7000	45	35
3/8-24	0.3750	0.0878			3620	23	17			5600	35	25	6900	45	45	7900	50	35
7/16-14	0.4375	0.1063			4380	30	24			6800	50	35	8400	60	45	9550	70	55
[illegible]	[illegible]	[illegible]			[illegible]	[illegible]	[illegible]			[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

[D44-53]

Estimated Results- Brakes

```
factor_of_safety = 1.9709

Applied torque: 1200 Nm
Friction torque: 2365.0888 Nm

The friction force is GREATER than the torque. The bolts can hold.
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Questions

Questions?